

BIB Week 2 — Biological data as files

- Dosya olarak biyolojik veri: referans sürümleri, koordinatlar ve sessiz hatalar
- Builds, coordinates, and silent failure
- Today you get the GENOMES card, and you keep it until Week 14
- Warning: Week 3 is code reading. This is the week to make sure your laptop works.

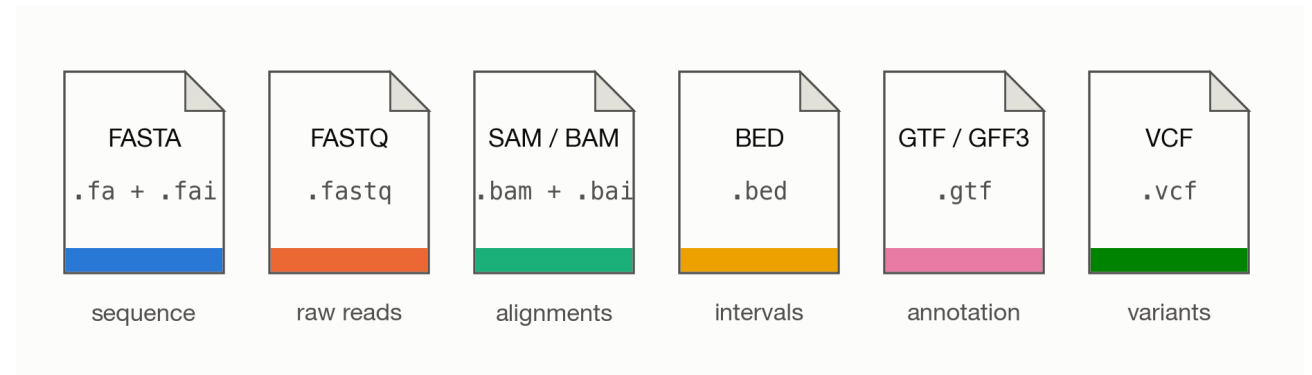
```
BED  chr17  7676520  7676594  TP53_exon  0  -  
GTF  chr17  HAVANA  exon  7676521  7676594  .  -  .
```

Ad of the Week #2

- This week: an advert that names a file format (dosya biçimi)
- They almost all do
- Tick your Skills Ledger

The six formats, fast

- FASTA (+ its .fai index)
- FASTQ
- SAM / BAM (and one word about CRAM)
- BED
- GTF / GFF3
- VCF
- For each: what it is for · what one record means · what it does not tell you



FASTA — and .fai

- For: sequence, and nothing else
- One record: a > header line, then sequence until the next >
- Does not tell you: the build, the species, the date, or whether it matches your BAM
- The .fai index lets a tool jump to chr17:7,500,000 without reading the file from the top
- The header line is free text. It is a promise from whoever made the file, not a fact.

```
>chr17
ACGTACGTAAGGCTTACAGGATCCAAGT...
>chr18
TTGACCAGGTTACGGATCCAAGTACGTA...
```

chr17-19.fa — GRCh38 chr17-19 subset, header written as '>chr17', 60 bases per line

[illegible]

file line 127,944 (sequence line 127,943; sed -n 127944p) begins at base 7,676,521 = 60 × 127,942 + 1 — the TP53 exon start
its byte offset = 7 ('>chr17' + newline) + 7,676,520 bases + 127,942 newlines = 7,804,469 (GENCODE's own '>chr17 17' header: 10)

chr17-19.fa.fai — one line per sequence, five columns

name	length	offset	linebases	linewidth	
chr17	83257441	7	60	61	offset = bytes before the first base
chr18	80373285	84645080	60	61	linewidth = 60 bases + 1 newline
chr19	58617616	166357927	60	61	that is all a tool needs to jump

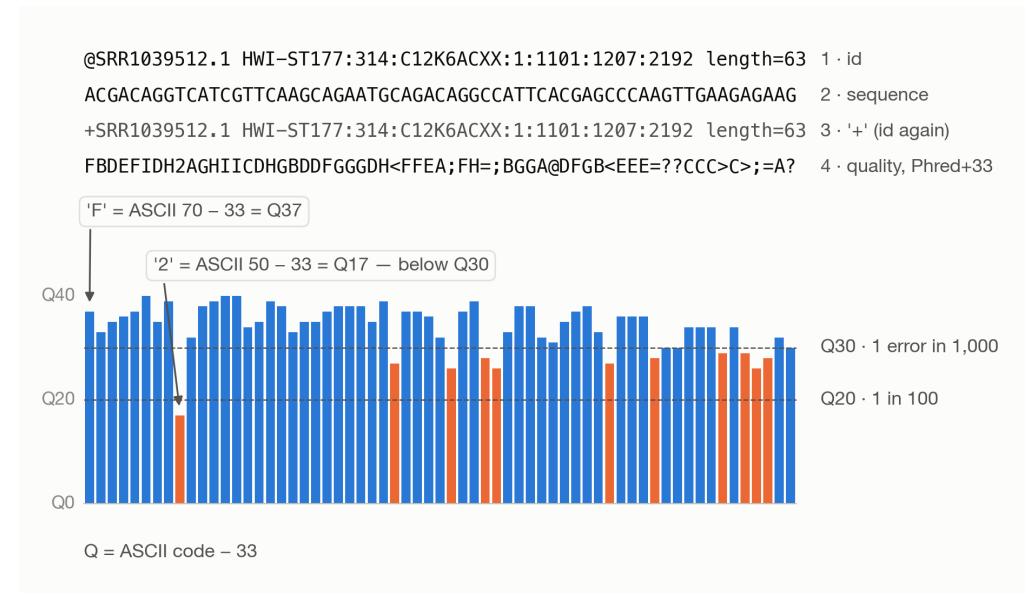
rows computed with the faidx arithmetic for this header style and line width; chr17/18/19 lengths are GRCh38

file lines 127,944-127,945 of chr17 are the real GRCh38 sequence at the TP53 exon start; .fai rows computed for a chr17-19 subset with '>chr17' headers at 60 bases/line

FASTQ

- For: raw reads, straight off the instrument
- One record: exactly four lines — id, sequence, +, quality
- Quality is Phred+33, and in 2026 that is the only encoding left in the wild
- $Q = -10 \cdot \log_{10}(P \text{ of error}) \rightarrow Q20 = 1 \text{ in } 100, Q30 = 1 \text{ in } 1000$
- Does not tell you: where the read came from in the genome. Nothing is aligned yet.

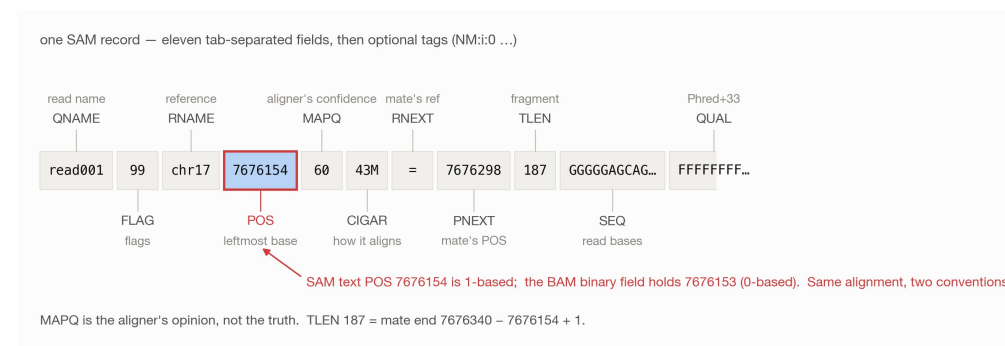
```
@A00519:512:HJ7CTDSX:1:1101:9878:1000 1:N:0:CGATGT
GGTTACAGGATCCAAGTACGTAGGCTTACAGGATCCAAGTACG
+
FFFFFF:FFFFFFFFFFFF,FFFFFFFF:FFFFF,FF:FFFFFF
```



First record of `example_file.fastq` (SRR1039512.1, 63 bp); bars are $Q = \text{ASCII} - 33$, orange below Q30

SAM / BAM — and one line on CRAM

- For: reads placed on a reference
- One record: one alignment of one read. POS is the leftmost mapped base.
- SAM is text; BAM is the same thing, compressed and indexed (.bai)
- SAM text POS is 1-based. The BAM binary POS field is 0-based. Same file, two conventions.
- CRAM is smaller still, but it is reference-dependent — lose the reference, lose the data
- Does not tell you: whether the alignment is correct. MAPQ is the aligner's own opinion.

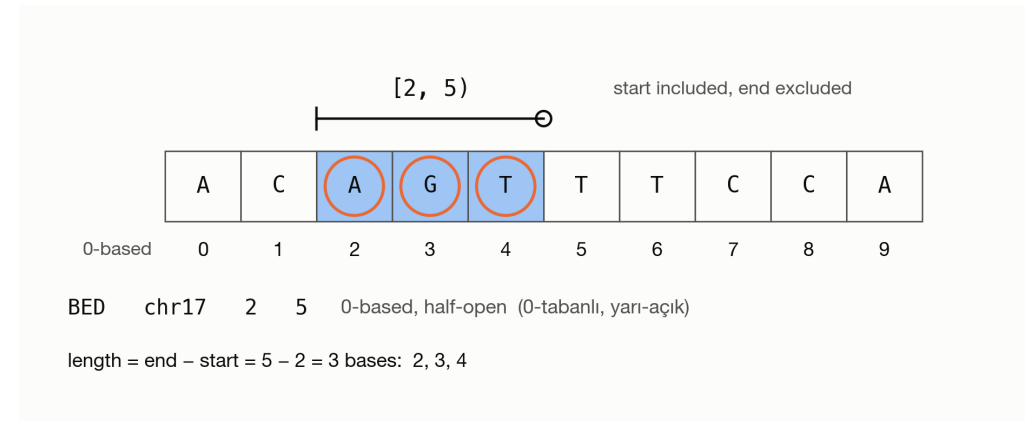


one SAM record, field by field; POS is 1-based in text, 0-based in the BAM binary

```
@SQ SN:chr1 LN:248956422
read001 99 chr17 7676154 60 43M = 7676298 187
GGGGGAGCAG... FFFFFFFFFF...
```

BED

- For: intervals. Peaks, exons, targets, anything with a start and an end.
- One record: one interval. Minimum three columns: chrom, start, end.
- 0-based, half-open. Length = end – start.
- The workhorse of bedtools, UCSC, and most peak callers
- Does not tell you: the build, the strand (unless column 6 is there), or what the interval means

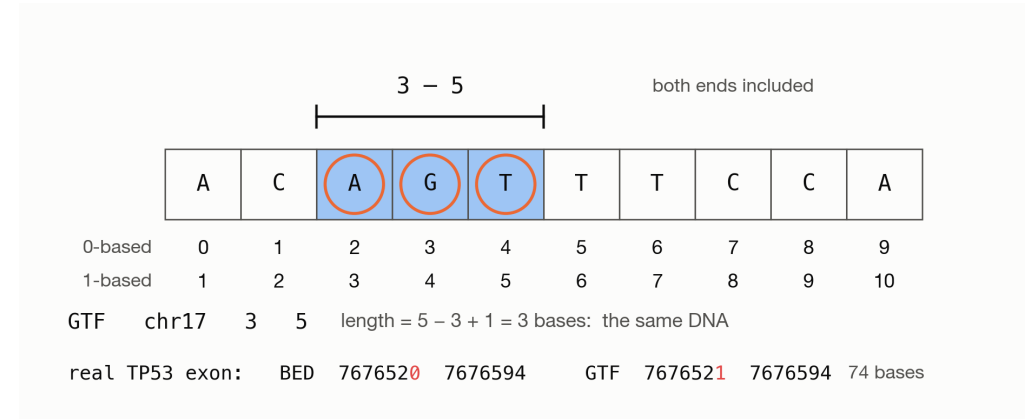


BED [2, 5): start included, end excluded, three bases

```
chr17 7668401 7687550 TP53 0 -
chr17 7676520 7676594 exon4 0 -
```

GTF / GFF3

- For: annotation. Where the genes, transcripts, exons and CDS are.
- One record: one feature, with a free-form attribute column at the end
- 1-based, inclusive. Length = end – start + 1.
- The attribute column is where the stable IDs live — ENSG..., ENST...
- Does not tell you: which release it is (unless the header comment survived) or which transcript to prefer



same ruler, 1-based: GTF 3-5; real TP53 exon BED 7676520-7676594
vs GTF 7676521-7676594

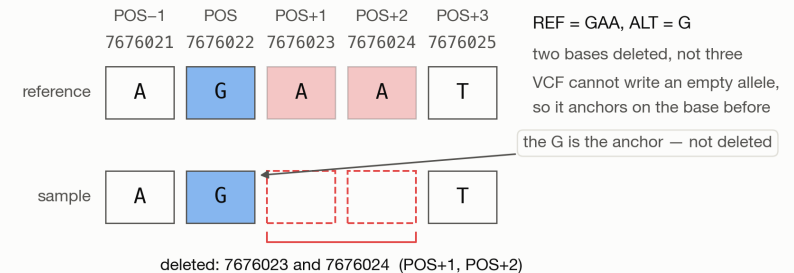
```
chr17 HAVANA exon 7676521 7676594 . - .  
gene_id "ENSG00000141510.19"; gene_name "TP53";
```

VCF

- For: variants — differences from the reference, per sample
- One record: one position, one or more alternate alleles, one column per sample
- 1-based, like GTF
- The indel anchor base: a deletion of AA is written as REF=GAA / ALT=G. The G is not deleted; it is an anchor.
- Does not tell you: whether the variant is real, or what it means

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	sample
chr17	7675088	.	C	T	832	PASS	DP=54	GT:DP	0/1:54 SNV (TP53 R175H)
chr17	7676022	.	GAA	G	611	PASS	DP=47	GT:DP	1/1:47 deletion

illustrative records: REF is the real GRCh38 base at each POS; QUAL, DP and GT are invented



POS names the anchor. The deletion starts at POS + 1. Code that treats POS as the first changed base is off by one on every indel.

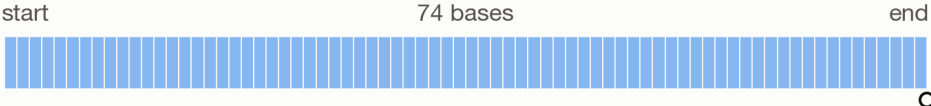
REF=GAA ALT=G at chr17:7676022 deletes 7676023-7676024; the G at POS is the anchor

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	sample
chr17	7675088	.	C	T	832	PASS	DP=54	GT:DP	0/1:54
chr17	7676022	.	GAA	G	611	PASS	DP=47	GT:DP	1/1:47

One feature, five ways — the numbers differ

- This one slide is the whole lesson.
Photograph it.

the same 74 bases of chromosome 17 (TP53 exon, minus strand) — written five ways



0-based, half-open 7676520 7676594

1-based, inclusive 7676521 7676594

BED	chr17	7676520	7676594	TP53_exon	0	-	end - start = 74		
GTF	chr17	HAVANA	exon	7676521	7676594	.	-	.	end - start + 1 = 74
VCF POS	chr17	7676521	.	A	G	.	PASS	SNV at the first base	
SAM text POS	read001	0	chr17	7676521	60	74M	BAM binary: 7676520		
Python slice	seq[7676520:7676594]							b - a = 74	

0-based, half-open: BED · BAM binary POS · Python slices · bedtools

1-based, inclusive: SAM text POS · VCF POS · GTF · GFF3 · GenBank · IGV · browser search boxes

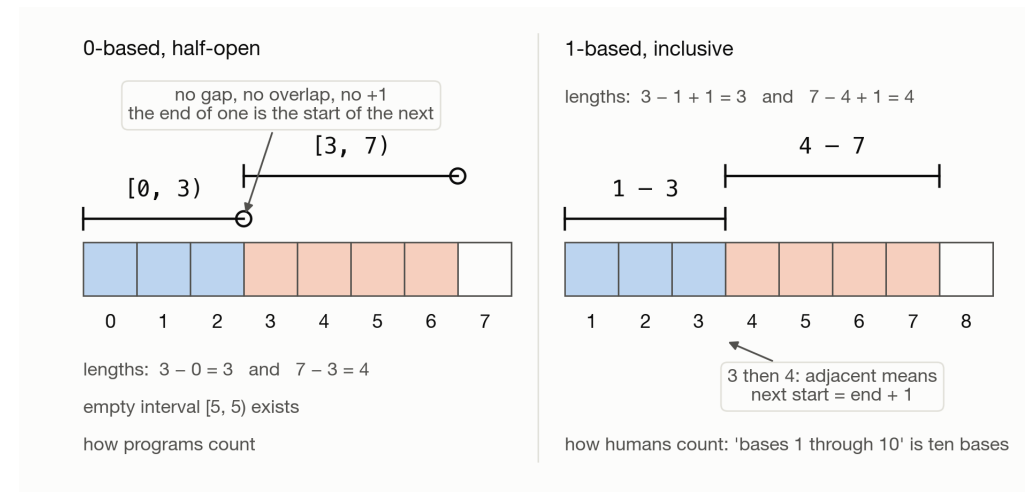
the red digit is the only difference: the start moves by one, the end does not, the length is 74 either way

Format	start	end	length
BED	7676520	7676594	end-start
GTF	7676521	7676594	end-start+1
VCF POS	7676521	—	—
SAM text POS	7676521	—	—
Python slice	7676520	7676594	b-a

one 74-base interval, five files, the differing digit in red

Why half-open is not perverse

- $[0, 3)$ and $[3, 7)$ are adjacent with no gap and no overlap
- Length is end – start. No +1. No off-by-one on every arithmetic step.
- An empty interval is expressible: $[5, 5)$. In 1-based inclusive, it is not.
- 1-based inclusive is how humans count: "bases 1 through 10" is ten bases
- Both are defensible. Neither is going away. That is the problem.



$[0,3)$ and $[3,7)$ touch with no +1; in 1-based the same intervals are 1-3 and 4-7

Worked on the board, twice

- Conversion 1 (BED $\rightarrow$ GTF): BED chr17 7676520 7676594 $\rightarrow$ GTF start +1 = 7676521, end unchanged
- Conversion 2 (GTF $\rightarrow$ BED): GTF 7676521 – 7676594 $\rightarrow$ BED start = 7676521 – 1 = 7676520; end unchanged
- The start moves. The end does not.
- Length: BED 7676594 – 7676520 = 74. GTF 7676594 – 7676521 + 1 = 74. Same DNA, same answer.
- If your two lengths disagree, you converted wrong. That is your check.

Now you do three, on paper

- A 1 bp feature. BED chr1 100 101. What is it in GTF? What is its length?
- A feature starting at the first base of the chromosome. GTF 1 – 10. What is it in BED?
- A deletion with an anchor base. VCF POS=1000 REF=GAA ALT=G — which bases are deleted (1-based)?
- Two minutes. Then swap with the person next to you and check theirs.
- Answers on the next slide. Do not turn round.

1. A 1 bp feature.

BED chr1 100 101

what is it in GTF? what is its length?

2. A feature at the first base of the chromosome.

GTF 1 – 10

what is it in BED?

3. A deletion with an anchor base.

VCF POS=1000 REF=GAA ALT=G

which bases are deleted, in 1-based coordinates?

start end length

1			
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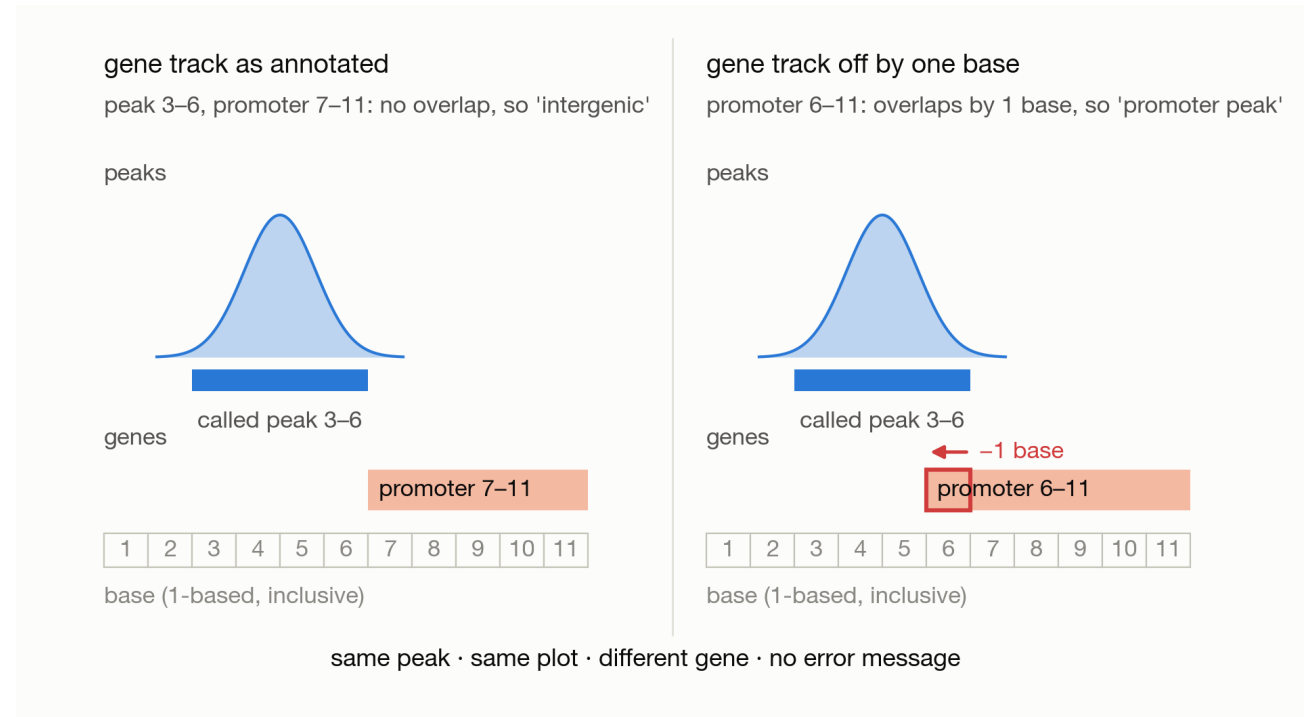
2			
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3			
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two minutes · then swap with your neighbour and look for a difference

Say the consequence out loud

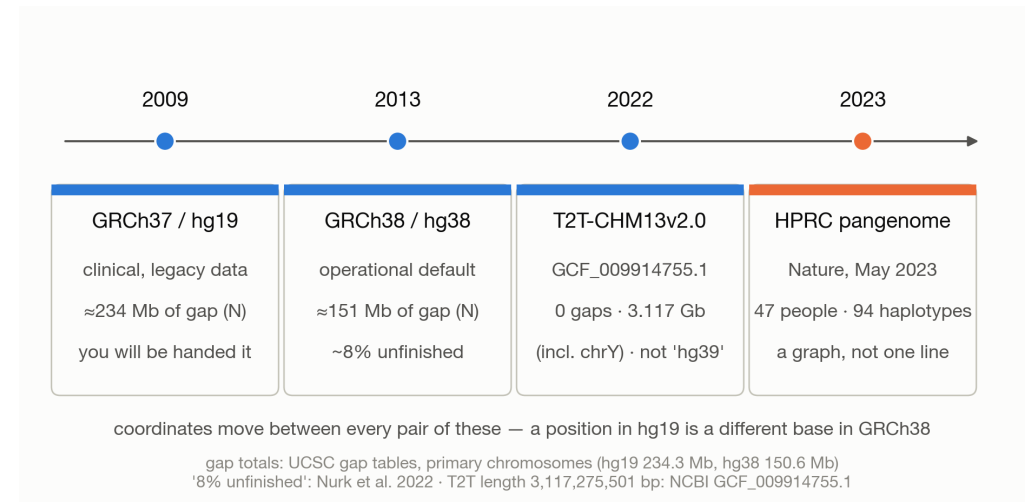
- An off-by-one runs cleanly
- It produces a plot
- The plot looks fine
- And it has quietly reassigned your peaks to the wrong genes
- Nobody will ever tell you. There is no error message for being one base wrong.



same peak; the promoter shifted by one base flips 'intergenic' to 'promoter peak'

The builds

- GRCh37 / hg19 — still dominant in clinical and legacy data. You will be handed hg19 files.
- GRCh38 / hg38 — the operational default since 2013
- T2T-CHM13v2.0 — gapless since 2022. Not "hg39". It is a different kind of thing.
- HPRC pangenome — the direction of travel: many genomes instead of one
- A build is not a version of a document. Coordinates move between builds.
- Turkish note: the population represented in these references is not this one — Week 13



Forensics: the file tells you which build

- It is in the @SQ header line of every SAM/BAM: @SQ SN:chr1 LN:248956422
- You do not need anybody to tell you. The file knows.
- This is the single most useful party trick in the course

chromosome 1 length	build
249,250,621	GRCh37 / hg19
248,956,422	GRCh38 / hg38
248,387,328	T2T-CHM13v2.0

GRCh37 / hg19 — chr1 is 249,250,621 bases

```
@HD VN:1.6 S0:coordinate
@SQ SN:chr1 LN:249250621 ← the fingerprint
... one @SQ line per contig
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem -t 8 hg19.fa r1.fq r2.fq
```

GRCh38 / hg38 — chr1 is 248,956,422 bases

```
@HD VN:1.6 S0:coordinate
@SQ SN:chr1 LN:248956422 ← the fingerprint
... one @SQ line per contig
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem -t 8 GRCh38.fa r1.fq r2.fq
```

T2T-CHM13v2.0 — chr1 is 248,387,328 bases

```
@HD VN:1.6 S0:coordinate
@SQ SN:chr1 LN:248387328 ← the fingerprint
... one @SQ line per contig
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem -t 8 chm13v2.0.fa r1.fq r2.fq
```

no two builds share a chr1 length; the file knows which build it is

chr1 length identifies the build: 249,250,621 / 248,956,422 / 248,387,328

The naming trap

- UCSC says chr1. Ensembl says 1. Same chromosome, different string.
- UCSC says chrM. Ensembl says MT.
- And in hg19, chrM is not even the same sequence as the rCRS mitochondrial reference
- chr17_GL000205v2_random, chrUn_KI270742v1, HLA-A*01:01:01:01 — the alt contigs will also ruin your day
- These are strings. String comparison does not know they mean the same thing.

what it is	UCSC / GENCODE style	Ensembl style
chromosome 1	chr1	1
chromosome 17	chr17	17
X chromosome	chrX	X
Y chromosome	chrY	Y
mitochondrion	chrM	MT
unplaced contig	chrUn_KI270742v1	KI270742.1
unlocalised on 17	chr17_GL000205v2_random	GL000205.2
MHC alt haplotype	chr6_GL000250v2_alt	HSCHR6_MHC_APD_CTG1
HLA allele (hs38DH)	HLA-A*01:01:01:01	— (absent)
EBV decoy (hs38DH)	chrEBV	— (absent)

the naming is not in the filename: GRCh38.primary_assembly.genome.fa (GENCODE) says chr17, Homo_sapiens.GRCh38.dna.primary_assembly.fa (Ensembl) says 17 — both say GRCh38, neither says which

*same entity, two strings; the filename says GRCh38 in both styles
and never says which*

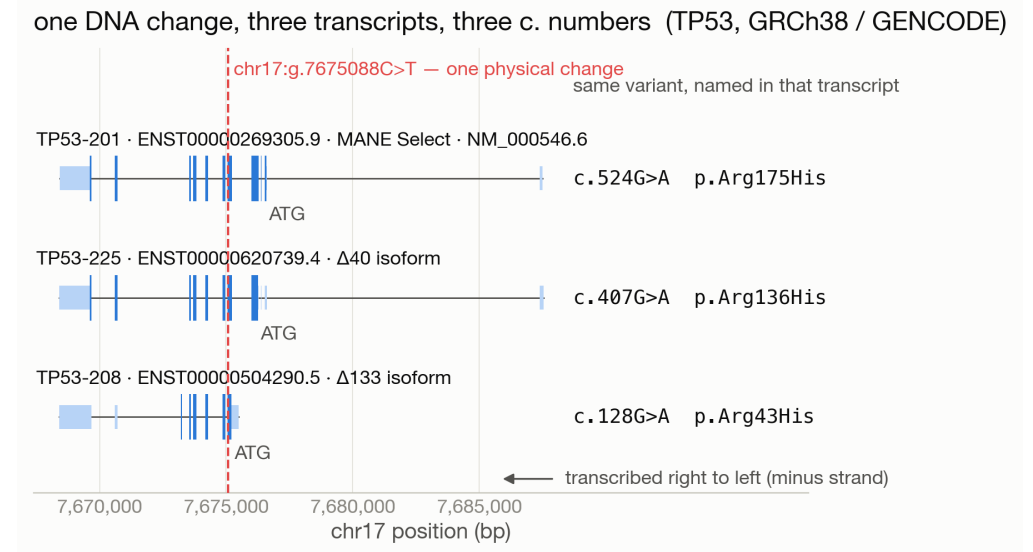
Silent failure, live

- Exit status 0. That means success.
- Output: zero rows.
- No warning. No error. No hint.
- liftOver between builds is lossy — intervals it cannot map are silently dropped
- And it is not re-calling: a lifted variant was never re-examined against the new reference

```
$ bedtools intersect -a peaks.chr.bed -b genes.nochr.gtf > out.bed
$ echo $?
0
$ wc -l out.bed
0 out.bed
```

The annotation release is part of the answer

- Build alone is not enough. GRCh38 + which GENCODE?
- GENCODE 50 (June 2026, GRCh38.p14) — the release we use all semester
- MANE Select v1.5 (March 2026): 19,437 transcripts covering 19,367 protein-coding genes
- MANE is the reason two labs on two continents now report the same variant the same way
- Ensembl 116 (June 2026) was the last legacy-platform release — old REST/FTP paths are archive-only



chr17:g.7675088C>T is c.524G>A (p.Arg175His) in the MANE transcript, c.407 in the delta-40 isoform, c.128 in the delta-133 isoform

The community renamed its genes because of a spreadsheet

- HGNC formally renamed 27 gene families in 2020
- SEPT2 → SEPTIN2 · MARCH1 → MARCHF1 · DEC1 → DELEC1
- Why: Excel silently autocorrected them to dates. SEPT2 → 2-Sep.
- Ziemann et al. 2016 found gene-name errors in ~20% of papers with supplementary Excel files
- The nomenclature committee changed reality to work around a spreadsheet default
- Bruford et al., Nat Genet 2020; <https://www.genenames.org/>

The GENOMES card

- G — Genome build and annotation release. Do all inputs agree?
- E — Environment, versions, and seed
- N — Numbers of rows and columns, before and after every filter
- O — Off-by-one and coordinate convention
- M — Multiple testing and the unit of independence — n/a — not yet taught until Week 6
- E — Entities resolved to stable IDs
- S — Sources resolved to a DOI, and opened

GENOMES	
seven controls · every notebook, every week	
your entry	
G	Genome build + annotation release: all inputs agree?
E	Environment: versions and the seed
N	Rows and columns, before and after every filter
O	Off-by-one: which coordinate convention?
M	Multiple testing and the unit of independence
E	Entities as stable IDs (ENSG..., not symbols)
S	Sources resolved to a DOI — and opened

BIB 2026 · Ege University · keep until Week 14

GENOMES	
yedi kontrol · her defter, her hafta	
girdiniz	
G	Genom derlemesi ve anotasyon sürümü: girdiler uyumlu mu?
E	Ortam: sürümler ve tohum (seed)
N	Satır/sütun sayıları, her filtreden önce ve sonra
O	Bir kayma (off-by-one): hangi koordinat kuralı?
M	Çoklu test ve bağımsızlık birimi
E	Varlıklar kararlı kimlikle (ENSG..., sembol değil)
S	Kaynaklar DOI'ye çözümlendi — ve açıldı

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One completed card, line by line

- Worked on last week's notebook — the one you already submitted
- G: GRCh38 / GENCODE 50 — and the GENCODE table shipped with the notebook is v50, so they agree
- E: Python 3.12, bib_colab 1.0, seed 48213
- N: 200 rows in the gene table; 1 row after selecting my gene
- O: the table is 1-based inclusive (it came from a GTF)
- M: n/a until W6
- E: ENSG00000141510.19, not "TP53"
- S: doi:10.1093/nar/gkae1078, opened and it is the GENCODE 2025 paper

GENOMES table as it renders in the notebook — one row per letter, M pre-filled

	row	entry — Notebook 1 (Week 1)
G	Build + annotation	GRCh38 · GENCODE 50 · shipped table is v50: agree
E	Environment	Python 3.12, bib_colab 1.0, seed 48213
N	Rows before / after	200 rows in gene table; 1 after selecting my gene
O	Off-by-one	1-based inclusive (the table came from a GTF)
M	Multiple testing	n/a until W6
E	Entities	ENSG00000141510.19 — not "TP53"
S	Sources	doi:10.1093/nar/gkae1078 · opened: GENCODE 2025

What is due — and a warning about next week

- Friday's lab: paper first, no laptops for the first nine minutes. Bring a pen.
- Lab Notebook 2 due Sunday 23:59. A.finish() is at minute 30 of the lab.
- The GENOMES table is required from this week on. Six rows. M says n/a until W6.
- Week 3 is reading Python. If your laptop has been marginal, fix it this week.
- Reading, 10 minutes, optional: hts-specs SAMv1.pdf §1.4, the POS field — link on the sources slide

BIB Lab 2 — predict, then check name _____ no. _____ seed 48213

your gene: TP53 · ENSG00000141510.19 · chr17, minus strand · one 74-base exon, printed five ways

								start	end	length
BED	chr17	7676520	7676594	TP53_exon	0	-				
GTF	chr17	HAVANA	exon	7676521	7676594	.	-			
VCF POS	chr17	7676521	.	A	G	.	PASS			
SAM text POS	read001	0	chr17	7676521	60	74M				
Python	seq[7676520:7676594]									

reconcile: do all five describe the same bases? length by BED _____ length by GTF _____

agree? ☐ yes ☐ no signature _____ minute _____

sign and hand in before any laptop opens.

Sources on this slide set

- SAM/BAM specification (POS, @SQ, CIGAR) — <https://samtools.github.io/hts-specs/SAMv1.pdf>
- VCF specification (POS, indel anchor base) — <https://samtools.github.io/hts-specs/VCFv4.3.pdf>
- UCSC coordinate FAQ (0-based half-open vs 1-based) — <https://genome.ucsc.edu/FAQ/FAQtracks.html#tracks1>
- GENCODE Release 50 — <https://www.gencodegenes.org/human/stats.html>
- MANE v1.5, March 2026 — <https://www.ncbi.nlm.nih.gov/refseq/MANE/>
- T2T-CHM13v2.0 (GCF_009914755.1); HPRC pangenome, Nature 2023
- Bruford et al., gene nomenclature guidelines, Nat Genet 2020; <https://www.genenames.org/>
- Ziemann, Eren & El-Osta, Genome Biol 2016 — doi:10.1186/s13059-016-1044-7
- Ensembl platform migration — <https://www.ensembl.info/2026/08/10/ensembl-is-moving-to-our-new-platform-this-week/>